

RUKMANGADA

+1-469-792-9738 ◇ Dallas, Texas

rukmandaruku@gmail.com ◇ <https://www.linkedin.com/in/rukmandada26/> ◇ <https://rukmandada26.github.io>

EXECUTIVE SUMMARY

- Aspiring Data Scientist with more than 5 years of experience. Proficient in translating business problems into data science problem, building machine learning models, developing ETL pipelines using PySpark on Airflow, building data visualization, performing data analysis & statistical tests.
- Skilled in developing advanced SQL queries, stored procedures, ETL processes, as well as optimizing queries & conducting performance tuning, expertise in MS SQL Server, MS Integration Services, MS Reporting Services, Big Query (GCP), Power BI, and Python

EDUCATION

Master of Science in Business Analytics

(GPA: 3.9) (Aug 2024 – Apr 2026)

University of Texas at Dallas

Bachelor of Technology in Electronics and Electrical Engineering

(GPA: 3.84) (2014 – 2018)

KIIT University, India

ACADEMIC PROJECTS

Multilabel Classification Project.

- Developed and evaluated five distinct deep learning approaches for multilabel emotion classification on Twitter data across 11 emotion categories. Implemented methodologies including:
 - Fine-tuning transformer based encoder models **ModernBERT and Gemma-2-2B with classification head using LoRA and 4-bit quantization**
 - Generative language model approach generating labels as JSON output with zero shot instruction-tuned LLM model with chat template format
 - Custom embedding Bag-based **MLP neural network**
- Optimized models using parameter-efficient fine-tuning (LoRA), gradient checkpointing, and 4-bit quantization, achieving efficient training on limited GPU resources while comparing performance across traditional, transformer-based, and generative architectures. **Deployed models for inference using vLLM with constrained decoding to ensure valid output formats.**

Credit Risk Prediction Model (Python, XGBoost, Neural Networks).

- Developed a credit risk prediction framework on the American Express dataset (5.5M rows, 190 features) using XGBoost and Neural Networks, achieving strong performance (XGBoost AUC approx 0.95 on holdout data).
- Engineered advanced behavioral features (averages, min/max, recency, rate-of-change) & reduced dimensionality to 17 high-value predictors via feature importance and SHAP analysis, improving accuracy and interpretability
- Optimized and validated models through Grid Search (72+ configurations), SHAP interpretability, and rank ordering, selecting XGBoost as the final model and designing conservative vs. aggressive approval strategies based on default thresholds.

Conagra Brands – Sales Forecasting and Consumer Insights(Sponsored by Conagra).

- Analyzed 50k+ retail transactions using clustering and LASSO regression, identifying key sales drivers (packaging, form factor, pricing) and uncovering three distinct consumer segments.
- Performed buyer persona profiling across demographics (income, household size, age, family status) to highlight high-value groups with significantly higher purchase likelihood.
- Delivered actionable growth strategies on product mix, distribution, and promotions; project was recognized as a Top 3 presentation out of 20% teams in class.

EXPERIENCE

Senior Data Engineer

Dec 2021–Jan 2024

Accenture Solutions Pvt. Ltd.

Mumbai, India

Project: Intelligent Insights to Business (I2B) – Migration to GCP

Successfully migrated Accenture's workforce analytics from MS SQL/SSIS → GCP (BigQuery, Airflow), delivering scalable, real-time, and predictive insights that improved hiring transparency, optimized workforce planning, and enhanced strategic decision-making.

- **Implemented forecasting models to predict headcount and attrition for future months**, giving leadership forward-looking insights for capacity planning and talent strategy.
- Collaborated with BI team to **develop dashboards with drill-down tiles** by Market, MU, DG, and DU, providing visibility into onboarding, attrition, and hybrid office attendance (swipe-in/out logs).
- Wrote optimized SQL queries in Google BigQuery to transform 10M+ monthly records into structured, analysis-ready datasets to improve the processing time over 20%.
- Automated reporting, reducing refresh times from 24 hours to less than 2 hours, enabling real-time decision-making

Data Engineer

Apr 2018–Nov 2021

Accenture Solutions Pvt. Ltd.

Mumbai, India

Built an Enterprise-wide People Analytics platform enabling leadership to track headcount, attrition, and recruitment metrics across different Org units like Markets, Market Units, Delivery Groups, and Delivery Units.

- Designed ETL workflows in MS SQL & SSIS to process 1M+ employee and recruitment records/month into analytics-ready tables.
- Developed parameterized stored procedures for role-based data access, providing secure insights to Market Leads, MU Leads, and DG/DU heads
- Collaborated with business stakeholders to analyze workforce data, identifying attrition trends and resource gaps, achieving a 10% reduction in production costs.

SKILLS

Languages and Database: Python, SQL, BigQuery, PySpark

Tools/Technologies: Natural Language Processing, Machine Learning, Deep Learning, Airflow, Power BI, Tableau, NumPy, Pandas, Scikit-Learn, Spacy, SciPy, Matplotlib, Linear Regression, Logistic Regression, Data Analysis and Visualization, Predictive Analytics, Statistics, Random Forest, XGBoost, PyTorch, Keras, TensorFlow, Causal Analytics, A/B testing

CERTIFICATIONS

- IIM Calcutta — Executive General Management Program
- Coursera: Meta Database Engineer

AWARDS

- Dean's Excellence Scholarship, UT Dallas (2024)
- Growth Catalyst Award, Accenture (2023)- For driving analytics adoption & cost savings during GCP migration.
- TechStar Award, Accenture (2022)- Ranked in top 1% of Accenture Technology talent (career levels 11–8).